

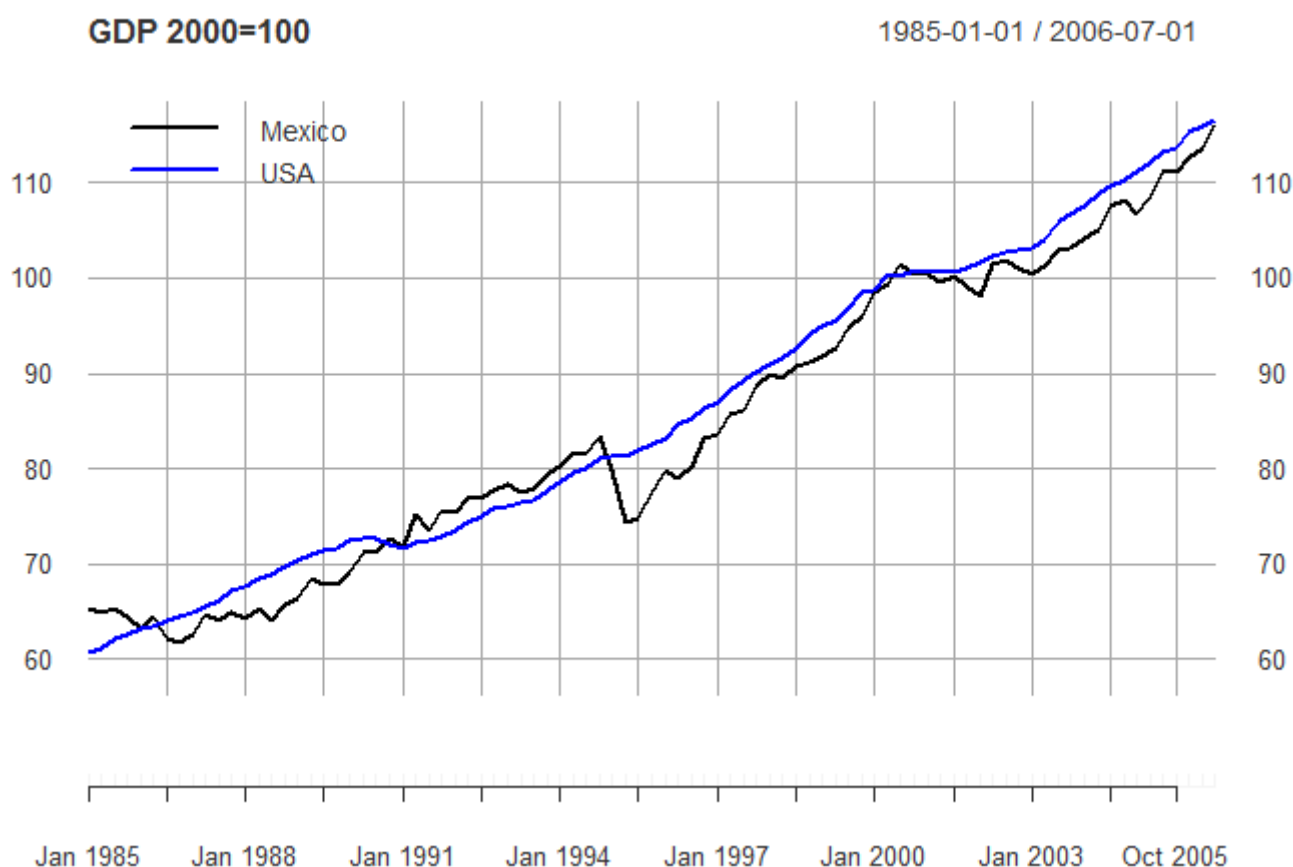
Advanced Econometrics – Lab 13 – Mock test

Exercise 1

This exercise is based on a computer exercise 12.10 from *Principles of Econometrics* textbook.¹

"The data file *mexico.dat* contains real GDP for Mexico and the United States from the first quarter of 1980 to the fourth quarter of 2006. Both series have been standardized so that the average value in 2000 is 100."

The plot below presents these two time series.



- (40%) Can both time series be cointegrated from an econometric perspective? Explain.
- (40%) Is there any economic reason that these two time series are cointegrated? Explain.
- (20%) PoP p. 345: "If the variables are not cointegrated, what should you do if you are interested in testing the relationship between Mexico and the United States?"

¹R. Carter Hill, William E. Griffiths and Guay C. Lim, *Principles of Econometrics*, 3rd Edition, Wiley.

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The output below presents stationarity analysis for Mexico.

```
> testdf2(variable = mexico$mexico, test.type = "c",
+         max.augmentations = 4, max.order = 4)
  augmentations      adf p_adf bgodfrey.stat bgodfrey.stat.1 bgodfrey.stat.2
1             0 1.0798783  0.99           86  8.798516e-02  0.117028675
2             1 1.0137292  0.99           85  1.191884e-06  0.072386158
3             2 1.0132774  0.99           84  3.857907e-04  0.004850674
4             3 1.1282872  0.99           83  2.039262e-02  0.020456585
5             4 0.6456936  0.99           82  3.909546e-02  0.039428133
  bg.stat.3 bg.stat.4 p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3 p_bg.p.value.4
1 2.1290622 4.8988383  0.7667544  0.9431647  0.5460557  0.2978357
2 2.1204758 5.0987492  0.9991289  0.9644541  0.5477815  0.2773145
3 2.0845120 4.9744383  0.9843293  0.9975776  0.5550524  0.2899304
4 0.2941272 2.8781597  0.8864460  0.9898238  0.9611292  0.5784177
5 0.5229714 0.5623214  0.8432597  0.9804790  0.9138165  0.9671572

> testdf2(variable = diff.xts(mexico$mexico), test.type = "nc",
+         max.augmentations = 4, max.order = 4)
  augmentations      adf      p_adf bgodfrey.stat bgodfrey.stat.1 bgodfrey.stat.2
1             0 -7.953195 0.01000000           85  1.56185215  1.5655972
2             1 -5.147982 0.01000000           84  0.68469911  2.4217807
3             2 -4.399151 0.01000000           83  0.91300600  2.5557925
4             3 -2.770934 0.01000000           82  0.26616752  0.4376153
5             4 -2.446282 0.01659992           81  0.07711433  0.2528479
  bg.stat.3 bg.stat.4 p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3 p_bg.p.value.4
1  3.971130  6.931061  0.2113942  0.4571249  0.2645985  0.13957650
2  5.284388  8.325938  0.4079733  0.2979319  0.1521187  0.08034245
3  5.986006  8.640352  0.3393181  0.2786228  0.1122930  0.07074528
4  2.175188  3.537417  0.6059143  0.8034762  0.5368520  0.47221181
5  2.018456  3.269774  0.7812468  0.8812412  0.5685850  0.51373547
```

- a) (40%) Is time series for Mexico stationary? Explain.
- b) (40%) Is time series for Mexico integrated of order 1? Explain.
- c) (20%) Can we use Mexico in cointegration analysis? Explain.

²R. Carter Hill, William E. Griffiths and Guay C. Lim, *Principles of Econometrics*, 3rd Edition, Wiley.

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p-value smaller than printed p-value

```
> # Estimating cointegrating vector
> model.coint <- lm(mexico~usa, data = mexico)
> summary(model.coint)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.65620    1.35300   1.963   0.0529 .
usa           0.95264    0.01548  61.549  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 2.351 on 85 degrees of freedom

Multiple R-squared: 0.9781, Adjusted R-squared: 0.9778

F-statistic: 3788 on 1 and 85 DF, p-value: < 2.2e-16

```
> # Testing stationarity of residuals
> testdf2(variable = residuals(model.coint), test.type="nc",
+         max.augmentations = 4, max.order = 4)
  augmentations      adf p_adf bgodfrey.statistic bgodfrey.statistic.1
1              0 -3.325962  0.01                86      2.492099e-02
2              1 -3.030915  0.01                85      1.164732e-05
3              2 -3.080097  0.01                84      4.522148e-03
4              3 -2.605995  0.01                83      4.141134e-02
5              4 -3.016641  0.01                82      1.134877e-02
  bgodfrey.statistic.2 bgodfrey.statistic.3 bgodfrey.statistic.4 p_bg.p.value.1
1          0.39810641          1.15432381          7.4576780      0.8745641
2          0.28090683          1.03090506          7.2332448      0.9972770
3          0.04966818          0.52464283          8.3336180      0.9463851
4          0.04778966          0.08688856          6.4795271      0.8387460
5          0.05011999          0.48417069          0.5726767      0.9151614
  p_bg.p.value.2 p_bg.p.value.3 p_bg.p.value.4
1          0.8195063          0.7639789      0.11359006
2          0.8689641          0.7937744      0.12406383
3          0.9754717          0.9134451      0.08009405
4          0.9763884          0.9933630      0.16608492
5          0.9752514          0.9223550      0.96605099
```

a) (50%) What is the cointegrating vector? How to interpret its elements?

b) (50%) Are Mexico and USA cointegrated? Explain.

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p-value smaller than printed p-value

```
> mexico$lagged_resid <- lag.xts(residuals(model.coint))
> # Estimating ECM
> model.ecm <- lm(diff.xts(mexico)~diff.xts(usa) + lagged_resid - 1,
+               # -1 means a model without a constant
+               data = mexico)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
diff.xts(usa)	0.85632	0.18726	4.573	1.64e-05	***
lagged_resid	-0.21164	0.06328	-3.344	0.00123	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.357 on 84 degrees of freedom
(1 observation deleted due to missingness)

Multiple R-squared: 0.2949, Adjusted R-squared: 0.2781

F-statistic: 17.56 on 2 and 84 DF, p-value: 4.24e-07

```
> bgtest(residuals(model.ecm)~1, order=1)
```

Breusch-Godfrey test for serial correlation of order up to 1

data: residuals(model.ecm) ~ 1

LM test = 0.0074, df = 1, p-value = 0.9314

```
> bgtest(residuals(model.ecm)~1, order=2)
```

Breusch-Godfrey test for serial correlation of order up to 2

data: residuals(model.ecm) ~ 1

LM test = 0.33815, df = 2, p-value = 0.8444

- a) (40%) Interpret parameters of the ECM model.
- b) (30%) Does ECM model work? Can we trust its estimates? Explain.
- c) (30%) How many lags of autocorrelation should we check for residuals? Explain.

⁴R. Carter Hill, William E. Griffiths and Guay C. Lim, *Principles of Econometrics*, 3rd Edition, Wiley.